

ADA-Marker Answer Sheet

Anonymized local demo

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PROBLEM

Q3

Answer

7. One vertex might be connected to multiple other vertex, and might also be connected from multiple other vertex.

In order to find the longest path, we can calculate the "path value" for each vertex. The "path value" can be calculated by $\max \{ \text{Path value of the } (i-1)\text{th vertex} \mid (i-1)\text{th vertex connect to the } i\text{th vertex} \} + 1$. By implementing this algorithm from the very origins down to the last vertex, we can obtain all of the path value of every vertex, and, thus, obtain the longest distance.

This algorithm ~~can~~ considers every possible connection between vertexes by viewing every edges, which takes $O(m)$.

On the other hand, this algorithm calculates the path value of every vertex, taking $O(n)$ time, with each calculation being linear time. In conclusion, it takes $O(n+m)$ time to obtain the longest path in an n -vertex m -edge directed acyclic graph.

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